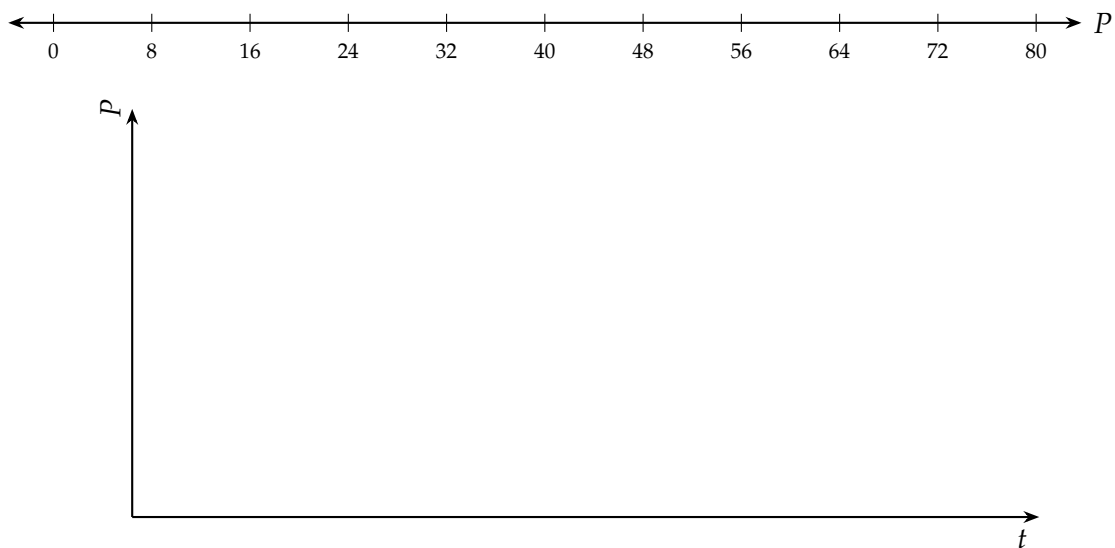


Lecture 13 Live Problems

Equilibria and parameters. The Allee effect.

Points toward lab manual pages: **Ph1, Ph2, Ph3, Ph4** (section L3 S2)

1. Logistic phase portrait. Given the change equation $P' = 0.25 P (1 - P/65)$, draw the phase portrait. Draw time-series plots too!



2. Solve by thought.

- (a) Solve the equation $1 - (X/3) = 0$ by thought.
- (b) Solve the equation $(X - 1)(X^2 + e^X) = 0$ by thought.

3. Critical mass: the Allee effect. Draw a phase portrait describing a population which only flourishes once it reaches a critical mass" of 10 individuals. The environment can only support 65 individuals. Then give a change equation which fits this phase portrait.

